Daniel Mihovch

732-239-5689 | dmihovch@udel.edu | linkedin.com/in/danielmihovch | github.com/dmihovch

Education

University of Delaware Newark, DE

Bachelor of Science in Computer Science, conc. Systems & Networks *Aug. 2023 – May 2027 (Expected)*

Experience

Qnity Electronics Wilmington, DE

Digital Technology Intern *May 2026 – Aug 2026*

- Created a Python data cleaning pipeline to reconcile company HR data with Copilot usage metrics, resolving errors across both datasets to form one cohesive, truthful dataset covering 6,000+ employees and contractors with over 99% accuracy, reconciling each user's management chain up to the CEO
- Developed multiple web-based dashboards for visualizing Copilot usage and adoption metrics across the organization
- Drove enterprise-wide AI adoption by running demo and training sessions for hundreds of employees globally, developing internal agents, holding 1:1 training sessions, and publishing company-wide communications on new features and use cases

FLIE Labs - Vertically Integrated Projects: Drone Newark, DE

Programmer *Feb 2026 – Present*

- Implementing autonomous swarming logic for DJI Matrice 100 drones utilizing ROS nodes running on a companion Raspberry Pi
- Developing a SITL to Hardware simulation pipeline to test high level drone swarming logic in a full software environment, and hardware simulation to test drone-specific flight controller logic
- Implementing a custom hybrid UDP networking stack to enable high speed peer to peer swarm communication, while guaranteeing delivery of vital messages

Four Youth Productions Wilmington, DE

STEM Instructor & Studio Operations Assistant *Feb 2025 – Present*

- Designed & led after-school STEM lessons (environmental science, physics, CS, chemistry) for 300+ underrepresented students across 4 Wilmington, DE public schools, grades Pre-K through 8th
- Administered company systems: Slack server, security system, NAS, and multiple Apple desktops
- Assisted in managing community events to showcase student work

University of Delaware CIS Newark, DE

Teaching Assistant *Aug 2024 – Present*

- TA for CISC106 (Python for Engineers), CISC210 (Systems Programming in C), and CISC260 (Computer Architecture & ARMv8 Assembly)
- Led weekly practicum sessions for 25+ students; held office hours and graded assignments
- Taught the C programming language, ARMv8 assembly, Bash, Unix systems, and Raspberry Pi hardware interaction
- Heavily assisted in running the CISC210 lecture section, including delivering a full lecture to the class

Projects

Piemulator - Raspberry Pi HAL | ***C, ASM, Bash, Raspberry Pi*** November 2025 – January 2026

- Created a Hardware Abstraction Layer that serves as a drop-in replacement for the University of Delaware's *libsense* Raspberry Pi Sense Hat library
- Focused on portability and ease of use, currently only depends on libc and pthreads. Earlier versions also depended on ncurses for terminal rendering, but this has since been simplified to ANSI escape codes
- Students can test and develop projects without always needing access to the physical Raspberry Pi, simply needed to link against my libpie instead of the provided libsense

DanOS - 16 bit Virtual Machine | ***C, Computer Architecture*** August 2025 – November 2025

- Single tasking, 16 bit virtual machine written in C
- Implemented my own RISC-style ISA featuring a segmented memory model and multiple addressing modes
- Currently writing a companion assembler to avoid having to write raw bytecode programs

Technical Skills

Languages: C, C++, Rust, Bash, Python
Operating Systems: Linux, OSX, Windows
Frameworks/Libraries: CUDA, OpenGL, Raylib, SDL, ncurses, React, Svelte, ROS, DJI API